

Set Language (9 th std)

Choose the Appropriate Answer

1. Which of the following is correct?

- (a) $\{7\} \in \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$ (b) $1 \in \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$
(c) $7 \notin \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$ (d) $\{7\} \not\subseteq \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$

Ans: (b) $1 \in \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$

2. The set $P = \{x \mid x \in \mathbb{Z}, -1 < x < 1\}$ is a

- (a) Singleton set (b) Power set (c) Null set (d) Subset

Ans: (a) Singleton set

3. If $U = \{x \mid x \in \mathbb{N}, x < 10\}$ and $A = \{x \mid x \in \mathbb{N}, 2 \leq x < 6\}$ then $(A')'$ is.....

- (a) $\{1, 6, 7, 8, 9\}$ (b) $\{1, 2, 3, 4\}$ (c) $\{2, 3, 4, 5\}$ (d) $\{\}$

Ans: (c) $\{2, 3, 4, 5\}$

$$\begin{aligned} U &= \{1, 2, 3, 4, 5, 6, 7, 8, 9\} \\ A &= \{2, 3, 4, 5\}; A' = \{1, 6, 7, 8, 9\} \\ (A')' &= U - A \\ (A')' &= \{2, 3, 4, 5\} \end{aligned}$$

4. If $B \subseteq A$ then $n(A \cap B)$ is.....

- (a) $n(A - B)$ (b) $n(B)$ (c) $n(B - A)$ (d) $n(A)$

Ans: (b) $n(B)$

5. If $A = \{x, y, z\}$ then the number of non-empty subsets of A is.....

- (a) 8 (b) 5 (c) 6 (d) 7

Ans: (d) 7

$$\begin{aligned} n(A) &= 3; P(A) = 2^m = 2^3 = 8 \\ \text{Non-empty subsets of } A &= 8 - 1 = 7 \end{aligned}$$

6. Which of the following is correct?

- (a) $\emptyset \subseteq \{a, b\}$ (b) $\emptyset \in \{a, b\}$ (c) $\{a\} \in \{a, b\}$ (d) $a \subseteq \{a, b\}$

Ans: (a) $\emptyset \subseteq \{a, b\}$

7. If $A \cup B = A \cap B$ then

- (a) $A \neq B$ (b) $A = B$ (c) $A \subset B$ (d) $B \subset A$

Ans: (b) $A = B$

8. If $B - A$ is B , then $A \cap B$ is.....

- (a) A (b) B (c) U (d) $\emptyset$

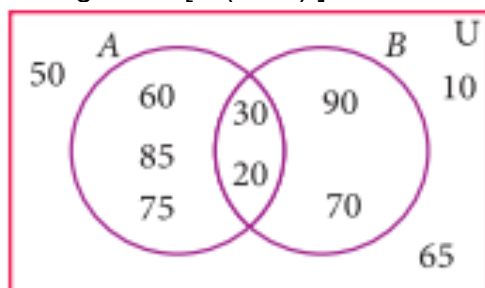
Ans: (d) $\emptyset$

9. Let $A = \{\emptyset\}$ and $B = P(A)$ then $A \cap B$ is.....

- (a) $\{\emptyset, \{\emptyset\}\}$ (b) $\{\emptyset\}$ (c) $\emptyset$ (d) $\{0\}$

Ans: (b) $\{\emptyset\}$

10. From the adjacent diagram $n[P(A \Delta B)]$ is.....



- (a) 8 (b) 16 (c) 32 (d) 64

Ans: (c) 32

$$\begin{aligned} A - B &= \{60, 85, 75\} & B - A &= \{90, 70\} \\ A \Delta B &= (A - B) \cup (B - A) = \{60, 70, 75, 85, 90\} \\ n[P(A \Delta B)] &= 2^5 = 32 \end{aligned}$$

11. If $n(A) = 10$ and $n(B) = 15$ then the minimum and maximum number of elements in $A \cap B$ is

- (a) (10, 15) (b) (15, 10) (c) (10, 0) (d) (0, 10)

Ans: (d) (0, 10)

12. In a class of 50 boys, 35 boys play carom and 20 boys play chess then the number of boys play both games is.....

- (a) 5 (b) 30 (c) 15 (d) 10

Ans: (a) 5

$$\begin{aligned} n(A \cup B) &= 50, n(A) = 35, n(B) = 20 \\ n(A \cap B) &= n(A) + n(B) - n(A \cup B) = 35 + 20 - 50 = 5 \end{aligned}$$

13. If $U = \{x : x \in N \text{ and } x < 10\}$, $A = \{1, 2, 3, 5, 8\}$ and $B = \{2, 5, 6, 7, 9\}$, then $n[(A \cup B)']$ is

- (a) 1 (b) 2 (c) 4 (d) 8

Ans: (a) 1

$$\begin{aligned} U &= \{1, 2, 3, 4, 5, 6, 7, 8, 9\} \\ A \cup B &= \{1, 2, 3, 5, 8\} \cup \{2, 5, 6, 7, 9\} = \{1, 2, 3, 5, 6, 7, 8, 9\} \\ (A \cup B)' &= \{1, 2, 3, 4, 5, 6, 7, 8, 9\} - \{1, 2, 3, 5, 6, 7, 8, 9\} = \{4\} \Rightarrow n(A \cup B)' = 1 \end{aligned}$$

14. For any three sets P, Q and R, $P - (Q \cap R)$ is

- (a) $P - (Q \cup R)$ (b) $(P \cap Q) - R$
(c) $(P - Q) \cup (P - R)$ (d) $(P - Q) \cap (P - R)$

Ans: (c) $(P - Q) \cup (P - R)$

15. Which of the following is true?

- (a) $A - B = A \cap B$ (b) $A - B = B - A$
(c) $(A \cup B)' = A' \cup B'$ (d) $(A \cap B)' = A' \cup B'$

Ans: (d) $(A \cap B)' = A' \cup B'$

16. If $n(A \cup B \cup C) = 100$, $n(A) = 4x$, $n(B) = 6x$, $n(C) = 5x$, $n(A \cap B) = 20$, $n(B \cap C) = 15$ and $n(A \cap C) = 25$, then $n(A \cap B \cap C) = 10$, then the value of x is.....
- (a) 10 (b) 15 (c) 25 (d) 30

Ans: (b) 15

$$n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C) - n(A \cap C) + n(A \cap B \cap C)$$

$$100 = 4x + 6x + 5x - 20 - 15 - 25 + 10$$

$$100 = 15x - 50 \Rightarrow 150 = 15x \Rightarrow x = 150 / 15 = 10$$

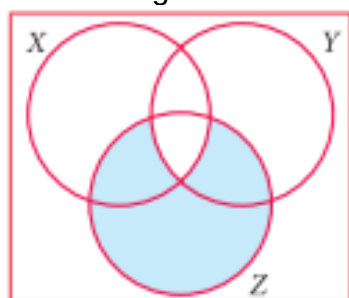
17. For any three sets A, B and C, $(A - B) \cap (B - C)$ is equal to
- (a) A only (b) B only (c) C only (d) $\emptyset$

Ans: (d) $\emptyset$

18. If J = Set of three sided shapes, K = Set of shapes with two equal sides and L = Set of shapes with right angle, then $J \cap K \cap L$ is.....
- (a) Set of isosceles triangles (b) Set of equilateral triangles
(c) Set of isosceles right triangles (d) Set of right angled triangles

Ans: (c) Set of isosceles right triangles

19. The shaded region in the Venn diagram is



- (a) $Z - (X \cup Y)$ (b) $(X \cup Y) \cap Z$ (c) $Z - (X \cap Y)$ (d) $Z \cup (X \cap Y)$
- Ans: (c) $Z - (X \cap Y)$**

20. In a city, 40% people like only one fruit, 35% people like only two fruits, 20% people like all the three fruits. How many percentage of people do not like any one of the above three fruits?
- (a) 5 (b) 8 (c) 10 (d) 15

Ans: (a) 5

21. Which of the following are well-defined sets?

- (a) All the colours in the rainbow
(b) All the points that lie on a curved line.
(c) All the honest members in the family
(d) All the efficient doctors of the hospital.

Ans: (a) All the colours in the rainbow

22. The set $A = \{x, x \in \mathbb{N}, \text{ and } x^2 - 3x + 2 = 0\}$ is

- (a) Null set (b) Finite set (c) Infinite set (d) None of these

Ans: (b) Finite set

23. The set $A = \{x, x \in \mathbb{R}, \text{ and } x^2 = 9, 2x = 4\}$ is
 (a) Empty set (b) Singleton set (c) Infinite set (d) None of these
Ans: (a) Empty set

24. Let $A = \{x: x \text{ is a letter in the word FOLLOW}\}$, $B = \{y: y \text{ is a letter in the word WOLF}\}$
 (a) A & B are disjoint (b) $A = B$ (c) $A \subset B$ (d) None of these
Ans: (b) $A = B$

25. Are the following pairs of sets equal?
 (a) $A = \{2\}$; $B = \{x: x \in \mathbb{N}, x \text{ is an even prime number}\}$.
 (b) $P = \{1, 4, 9\}$; $Q = \{x: x = n^2, n \in \mathbb{N}, n \leq 6\}$
 (c) $X = \{x: x \in \mathbb{W}, x < 5\}$; $Y = \{x: x \in \mathbb{N}, x \leq 5\}$
 (d) $M = \{a, b, c, d\}$; $N = \{p, q, r, s\}$
Ans: (a) $A = \{2\}$; $B = \{x: x \in \mathbb{N}, x \text{ is an even prime number}\}$

26. Which of the following are not equivalent sets?
 (a) $A = \{1, 2, 3\}$; $B = \{4, 5\}$
 (b) $P = \{q, s, m\}$; $Q = \{6, 9, 12\}$
 (c) $X = \{x: x \text{ is a prime number less than } 10\}$; $Y = \{x: x \in \mathbb{N}, x \leq 4\}$
 (d) $R = \{x: x = 2n + 3, n < 4, n \in \mathbb{N}\}$; $S = \{x: x = n / (n + 2), n \in \mathbb{R}, n \leq 5\}$
Ans: $A = \{1, 2, 3\}$; $B = \{4, 5\}$

27. If $A \cap B^c = \emptyset$
 (a) $A = B$ (b) $B \neq A$ (c) A is proper subset of B (d) None of these
Ans: (c) A is proper subset of B

28. $A^c - B^c$ is equal to
 (a) $B - A$ (b) $A - B$ (c) $A = B$ (d) None of these
Ans: (a) $B - A$

29. If $A = \emptyset$ then total number of elements in $P(A)$ are
 (a) No element (b) Zero (c) Two (d) One
Ans: (d) One

30. Let $A = \{a, b, c\}$ and $B = \{1, 2\}$ then the number of relations from A into B are
 (a) 6 (b) 5 (c) 32 (d) 64
Ans: (d) 64

31. Let R is the set of all triangles in a plane a R b if a is congruent to b, then R is
 (a) Only reflexive (b) Only Symmetric
 (c) Only Transitive relation (d) Equivalence relation
Ans: (d) Equivalence relation

32. The relation "is parallel" on the set A of all coplanar straight line is :
 (a) Only reflexive (b) Only Symmetric
 (c) Only Transitive relation (d) Equivalence relation
Ans: (d) Equivalence relation

33.If A and B are two sets, then $A \cup B$ contains

- (a) Only elements common on both (b) All elements of A and B
(c) Only elements of A (d) Only elements of B

Ans: (b) All elements of A and B

34.The compliment of universal set is _____

- (a) Itself (b) Empty set (c) Singleton set (d) None of these

Ans: (b) Empty set

35.A set which contains only one element is called _____

- (a) Null set (b) Singleton set (c) Finite set (d) Empty set

Ans: (b) Singleton set

36.Let $A = \{a, b, c\}$ and $R = \{(b, b), (c, a), (a, c)\}$, then the relation R on A is

- (a) Only reflexive (b) Only Symmetric
(c) Only Transitive relation (d) None of these

Ans: (b) Only Symmetric

37.The relation “congruence modulo m” is

- (a) An equivalence (b) Reflexive only
(c) Symmetric only (d) Transitive only

Ans: (a) An equivalence

38.If $a \in \mathbb{N} = \{a \mid x \in \mathbb{N}\}$ then the set $3\mathbb{N} \cap 7\mathbb{N}$ is equal to

- (a) $7\mathbb{N}$ (b) $3\mathbb{N}$ (c) $21\mathbb{N}$ (d) $\emptyset$

Ans: (c) $21\mathbb{N}$

39.A set has n elements, then the total number of subsets are

- (a) 2^n (b) 2^{n-1} (c) 2^{2n} (d) None of these

Ans: (a) 2^n

40.A set has n elements, then the total number of proper subsets are

- (a) 2^n (b) 2^{n-1} (c) 2^{2n} (d) None of these

Ans: (b) 2^{n-1}

41.The sets A & B have 6 & 9 elements respectively, such that A is proper subset B, then the total number of elements $A \cap B$ are

- (a) 6 (b) 9 (c) 3 (d) 15

Ans: (a) 6

42.The sets A & B have 5 & 9 elements respectively, such that A is proper subset B, then the total number of elements $A \cup B$ are

- (a) 5 (b) 9 (c) 14 (d) 4

Ans: (b) 9

43.The smallest set A such that $A \cup \{4, 5\} = \{1, 2, 3, 4, 5\}$ is

- (a) $\{3, 4, 5\}$ (b) $\{1, 2, 3\}$ (c) $\{1, 2\}$ (d) $\{1, 2, 3, 4, 5\}$

Ans: (b) $\{1, 2, 3\}$

44. Let X is a finite set containing n distinct elements, then total number of relation on X are equal to

- (a) 2^n (b) 2^{n-1} (c) 2^{2n} (d) 2^{n^2}

Ans: (d) 2^{n^2}

45. Which set is the subsets of all given sets

- (a) $\{1\}$ (b) $\{0\}$ (c) $\emptyset$ (d) $\{0, 1, 6, 7\}$

Ans: (c) $\emptyset$

46. If $A = \{1, 2, 3\}$ & $B = \{4, 5, 6\}$ then, $n(A \times B)$ is equal to

- (a) 6 (b) 9 (c) 27 (d) None of these

Ans: (b) 9

47. The number of relation that can be defined on the set $A = \{a, b, c\}$ are

- (a) 2^9 (b) 2^3 (c) 9^2 (d) 9

Ans: (a) 2^9

48. Let $X = \{1, 2, 3\}$ then the relation $R = \{(1,1), (2,2), (3,1)\}$ on X is

- (a) Reflexive (b) Symmetric (c) Transitive (d) None of these

Ans: (d) None of these

49. Let X & Y are two finite sets such that $O(X) = m$ & $O(Y) = n$ then the number of relations from X to Y are

- (a) 2^{m+n} (b) $m + n$ (c) $m n$ (d) 2^{mn}

Ans: (d) 2^{mn}

50. If A & B are two sets such that $n(A) = 15$, $n(B) = 21$, & $n(A \cup B) = 36$ then $n(A \cap B)$ equal to

- (a) 2 (b) 0 (c) 4 (d) 15

Ans: (b) 0

51. If P & Q are two sets such that $P \cup Q$ has 20 elements, P has 9 elements & Q has 16 elements. How many elements does $P \cap Q$ have?

- (a) 5 (b) 4 (c) 3 (d) 0

Ans: (a) 5

52. In a Group of 300 people, 150 can speak French & 200 can speak German. How many can speak both French & German?

- (a) 40 (b) 50 (c) 20 (d) None of these

Ans: (b) 50

53. The relation R defined on the set of natural numbers as $\{(a, b): a \text{ differs from } b \text{ by } 3\}$ is given

- (a) $\{(1, 4), (2, 5), (3, 6), \dots\}$ (b) $\{(4, 1), (5, 2), (6, 3), \dots\}$
(c) $\{(4, 1), (5, 2), (6, 3), \dots\}$ (d) None of the above

Ans: (d) None of the above

54. What is the possible number of reflexive relations on a set of 5 elements

- (a) 2^{10} (b) 2^{15} (c) 2^{20} (d) 2^{25}

Ans: (c) 2^{20}

55. The relation R defined on the set $A = \{1, 2, 3, 4, 5\}$ by $R = \{(x, y) : |x^2 - y^2| < 16\}$ is given by

- (a) $\{(1, 1), (2, 1), (3, 1), (4, 1), (2, 3)\}$ (b) $\{(2, 2), (3, 2), (4, 2), (2, 4)\}$
 (c) $\{(3, 3), (4, 3), (5, 4), (3, 4)\}$ (d) None of the above

Ans: (d) None of the above

56. If $A = \{1, 2, 3, 4\}$. Let $\sim = \{(1, 2), (1, 3), (4, 2)\}$. Then $\sim$ is

- (a) Not anti – symmetric (b) Transitive (c) Reflexive (d) Symmetric

Ans: (a) Not anti – symmetric

57. If $R = \{(1, 2), (2, 3), (3, 3)\}$ be a relation defined on $A = \{1, 2, 3\}$ then $R = R^2$ is

- (a) R itself (b) $\{(1, 2), (2, 3), (3, 3)\}$
 (c) $\{(1, 3), (2, 3), (3, 3)\}$ (d) $\{(2, 1), (1, 3), (2, 3)\}$

Ans: (c) $\{(1, 3), (2, 3), (3, 3)\}$

58. How many onto (surjective) functions are there from an n -element ($n \geq 2$) set to a 2-element set?

- (a) 2^n (b) 2^{n-1} (c) 2^{n-2} (d) $2^{(2n-2)}$

Ans: (c) 2^{n-2}

59. Consider the binary relation $R = \{(x, y), (x, z), (z, x), (z, y)\}$ on the set $\{x, y, z\}$, which one of the following is true?

- (a) R is symmetric but Not anti – symmetric
 (b) R is not symmetric but anti – symmetric
 (c) R is both symmetric and anti – symmetric
 (d) R is neither symmetric nor anti – symmetric

Ans: (d) R is neither symmetric nor anti – symmetric

60. For a set A , the power set of A is denoted by 2^A . If $A = \{5, 6, 7\}$, which of the following options are true?

- (a) $5 \in 2^A$ (b) $6 \subseteq 2^A$ (c) $\{5, 6\} \in 2^A$ (d) $\{5, 6\} \subseteq 2^A$

Ans: (c) $\{5, 6\} \in 2^A$

61. If f is a function from A to B , where $O(A) = m$ & $O(B) = n$, then total number of distinct functions are

- (a) n^m (b) n^m (c) m^n (d) $m + n$

Ans: (b) n^m

62. A function f from N to N defined by $f(n) = 2n+5 \in$ is

- (a) Many – one function (b) Into function
 (c) Onto function (d) Bijective function

Ans: (b) Into function

63. The number of binary relation on a set with n elements is

- (a) n^2 (b) 2^n (c) 2^{n^2} (d) None of these

Ans: (c) 2^{n^2}

64. The number of equivalence relations of the set $\{1, 2, 3, 4\}$ is

- (a) 4 (b) 15 (c) 16 (d) 24

Ans: (b) 15

65. Let A be a finite set of size n , the number of elements in the power set of $A \times A$ is
(a) 2^{2n} (b) $2n^2$ (c) 2^n (d) None of these

Ans: (b) $2n^2$

66. Which of the following set(s) are empty?

- (a) $\{x: x = x\}$ (b) $\{x: x \neq x\}$ (c) $\{x: x = x^2\}$ (d) $\{x: x \neq x^2\}$

Ans: (b) $\{x: x \neq x\}$

67. If $A = \{x, y\}$, the power set of A is

- (a) $\{\{x\}, \{y\}\}$ (b) $\{\{\emptyset\}, \{x, y\}\}$ (c) $\{\emptyset, \{x\}, \{y\}\}$ (d) None of these

Ans: (d) None of these

68. If A & B are sets and $A \cap B = A \cup B$, then

- (a) $A = \emptyset$ (b) $B = \emptyset$ (c) $A = B$ (d) None of these

Ans: (c) $A = B$

69. The domain & range are same for

- (a) Constant function (b) Identity function
(c) Absolute value function (d) Greatest integer function

Ans: (b) Identity function

70. Set A has 3 elements & set B has 4 elements. The number of injections that can be defined from A into B

- (a) 144 (b) 12 (c) 24 (d) 64

Ans: (c) 24

71. The number of bijective functions from set A to itself when A contains 106 elements is

- (a) 106 (b) 106^2 (c) $106!$ (d) 2^{106}

Ans: (c) $106!$

72. Let Z denote the set of all integers define f by $f(x) = x/2$, if x is even x , if x is odd then f is

- (a) Onto but not one-one (b) One-one but not onto
(c) One-one & onto (d) Neither one-one nor onto

Ans: (a) Onto but not one-one

73. To have inverse for the function f , f is

- (a) One – one (b) Onto (c) One – one onto (d) Identity function

Ans: (c) One – one onto

74. If $[x]$ denotes integral part of the real number, then the function $f(x) = x - [x]$ is a/an

- (a) Even function (b) Odd function (c) Periodic function (d) Constant

Ans: (c) Periodic function

75. The function $f: Z \rightarrow Z$ given by $f(x) = x^2$ is

- (a) One – one (b) Onto (c) One – one & onto (d) None of these

Ans: (d) None of these

76. Let $A = \{x: -1 < x < 1\} = B$. The function $f(x) = x/2$ from A to B is
 (a) Injective (b) Surjective
 (c) Both Injective & Surjective (d) Neither Injective nor Surjective

Ans: (a) Injective

77. $A - (B \cup C)$ is equal to
 (a) $(A - B) \cup (A - C)$ (b) $A - B - C$ (c) $(A - B) \cap (A - C)$ (d) $(A - B) \cup C$
Ans: (c) $(A - B) \cap (A - C)$

78. Which of the following is a set?
 (a) A collection of honest people.
 (b) A collection of hardworking students.
 (c) A collection of all natural numbers less than 10.
 (d) A collection of easy questions in this book.
Ans: (c) A collection of all natural numbers less than 10

79. If $A = \{1, 2, 3, 4, 5\}$, then the number of subsets of A is:
 (a) 5 (b) 10 (c) 25 (d) 32

Ans: (d) 32

$$2^n = 2^5 = 32$$

80. The set of all letters in the word "MATHEMATICS" in Roster form is:
 (a) $\{M, A, T, H, E, M, A, T, I, C, S\}$ (b) $\{M, A, T, H, E, I, C, S\}$
 (c) $\{M, A, T, H, E\}$ (d) $\{M, A, T, H, E, M, T, I, C, S\}$
Ans: (b) $\{M, A, T, H, E, I, C, S\}$

81. $A = \{1, 2, 3, 4, 5, 6\}$ and $B = \{4, 5, 6, 7, 8\}$, then $A \cap B$ is:
 (a) $\{1, 2, 3\}$ (b) $\{7, 8\}$ (c) $\{4, 5, 6\}$ (d) $\{1, 2, 3, 4, 5, 6, 7, 8\}$
Ans: (c) $\{4, 5, 6\}$

82. If $U = \{1, 2, 3, \dots, 10\}$ and $A = \{1, 3, 5, 7, 9\}$, then A' (Complement of A) is:
 (a) $\{1, 3, 5, 7, 9\}$ (b) $\{2, 4, 6, 8, 10\}$
 (c) $\{1, 2, 3, 4, 5\}$ (d) $\{6, 7, 8, 9, 10\}$
Ans: (b) $\{2, 4, 6, 8, 10\}$

83. If A , B and C are three sets, then $A - (B - C)$ equals to
 (a) $A - (B \cap C)$ (b) $(A - B) \cup C$
 (c) $(A - B) \cup (A \cap C)$ (d) $(A - B) \cup (A - C)$
Ans: (c) $(A - B) \cup (A \cap C)$

84. How many rational and irrational numbers are possible between 0 and 1?
 (a) 0 (b) Finite (c) Infinite (d) 1
Ans: (c) Infinite

85. Empty set is a -----?
 (a) Finite Set (b) Invalid Set (c) None of the above (d) Infinite Set
Ans: (a) Finite Set

86.If A,B and C are any three sets, then $A \cap (B \Delta C) =$

- (a) $(A \cap B) \Delta (A \cap C)$ (b) $(A \cap B) \Delta (A - C)$
 (c) $(A + B) \Delta (A \cap C)$ (d) Both a and b

Ans: (a) $(A \cap B) \Delta (A \cap C)$

87.If $A = [5, 6, 7]$ and $B = [7, 8, 9]$ then $A \cup B$ is equal to

- (a) $\{5, 6, 7, 8, 9\}$ (b) $\{5, 6, 7\}$ (c) $\{7, 8, 9\}$ (d) None of these

Ans: $\{5, 6, 7, 8, 9\}$

88.If $R = \{(1, -1), (2, -2), (3, -1)\}$ is relation, then find the range of R.

- (a) $\{-1, -2, -1\}$ (b) $\{1, -2, -1\}$ (c) $\{-1, 2, -1\}$ (d) None of these

Ans: (a) $\{-1, -2, -1\}$

89.The members of the set $S = \{x \mid x \text{ is the square of an integer and } x < 100\}$ is

- (a) $\{0, 2, 4, 5, 9, 58, 49, 56, 99, 12\}$ (b) $\{0, 1, 4, 9, 16, 25, 36, 49, 64, 81\}$
 (c) $\{1, 4, 9, 16, 25, 36, 64, 81, 85, 99\}$ (d) $\{0, 1, 4, 9, 16, 25, 36, 49, 64, \text{ and } 121\}$

Ans: (b) $\{0, 1, 4, 9, 16, 25, 36, 49, 64, 81\}$

90.Two finite sets have N and M elements. The number of elements in the power set of first set is 48 more than the total number of elements in power set of the second test. Then the value of M and N are

- (a) 7, 6 (b) 6, 4 (c) 7, 4 (d) 6, 3

Ans: (b) 6, 4

Number of subsets of B = 2^m Now, according to question $2^m - 2^n = 48$
 $\Rightarrow 2^n (2^m - n - 1) = 24(22 - 1)$ So, $n = 4$ and $m - n = 2$
 $\Rightarrow m - 4 = 2 \Rightarrow m = 2 + 4 \Rightarrow m = 6$

91.If A,B and C are three sets such that $A \cap B = A \cap C$ and $A \cup B = A \cup C$, then

- (a) $A = B$ (b) $A = C$ (c) $B = C$ (d) $A \cap B = \phi$

Ans: (c) $B = C$

92. $A - B$ is read as?

- (a) Difference of A and B of B and A (b) Both a and b
 (c) Difference of B and A (d) None of the above

Ans: (a) Difference of A and B of B and A

93.Which of the following two sets are equal?

- (a) $A = \{1, 2\}$ and $B = \{1\}$ (b) $A = \{1, 2\}$ and $B = \{1, 2, 3\}$
 (c) $A = \{1, 2, 3\}$ and $B = \{2, 1, 3\}$ (d) $A = \{1, 2, 4\}$ and $B = \{1, 2, 3\}$

Ans: (c) $A = \{1, 2, 3\}$ and $B = \{2, 1, 3\}$

94.In a class of 50 students 35 opted for Mathematics and 37 opted for Biology How may have opted for only Mathematics?

- (a) 15 (b) 17 (c) 13 (d) 19

Ans: (c) 13

$$n(M \cap B) = n(M) + n(B) - n(M \cup B) = 35 + 37 - 50 = 22$$

95. If $n(A) = 3$ and $n(B) = 6$ and $A \subseteq B$, then the number of elements in $A \cap B$ is equal to
(a) 3 (b) 9 (c) 6 (d) None of these

Ans: (a) 3

96. Let $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$, $A = \{1, 2, 5\}$, $B = \{6, 7\}$ then $A \cap B'$ is
(a) B' (b) A (c) A' (d) B

Ans: (b) A

97. If $(x - 1, y + 1) = (5, 6)$, then the value of x and y is given by
(a) $x = 5, y = 5$ (b) $x = 6, y = 5$ (c) $x = 5, y = 6$ (d) $x = 0, y = 0$

Ans: (b) $x = 6, y = 5$

98. What is the Cardinality of the Power set of the set $\{0, 1, 2\}$ =?
(a) 8 (b) 6 (c) 7 (d) 9

Ans: (a) 8

$$P(\{0, 1, 2\}) = \{\text{null}, \{0\}, \{1\}, \{2\}, \{0, 1\}, \{0, 2\}, \{1, 2\}, \{0, 1, 2\}\}.$$

99. Let A and B be two non-empty subsets of a set X such that A is not a subset of B , then

- (a) A is always a subset of the complement of B
- (b) B is always a subset of A
- (c) A and B are always disjoint
- (d) A and the complement of B are always non disjoint

Ans: (d) A and the complement of B are always non - disjoint

100. If ' A ' and ' B ' are finite sets, then which one of the following is the correct equation?

- (a) $n(A - B) = n(A) - n(B)$
- (b) $n(A - B) = n(B - A)$
- (c) $n(A - B) = n(A) - n(A \cap B)$
- (d) $n(A - B) = n(B) - n(A \cap B)$

Ans: (c) $n(A - B) = n(A) - n(A \cap B)$